

Berk Yağız KALAFAT

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Education

Istanbul Atlas University

Bachelor of Computer Science

Thesis: Privacy-Preserving Credit Card Fraud Detection via Federated Learning

Istanbul, Türkiye

Expected May 2027

Technical Projects

Privacy-Preserving Fraud Detection System (Graduation Project)

October 2025 – June 2026

- Architected an end-to-end MLOps pipeline using Federated Learning (flwr) to train distributed fraud detection models, ensuring 100% data privacy and strict GDPR compliance.
- Resolved extreme class imbalance by integrating the SMOTE algorithm at the local client level prior to decentralized training.
- Containerized the system with Docker, tracked training metrics via MLflow, and deployed the global model as a RESTful microservice using FastAPI.

Artificial Neural Network (ANN) Engine from Scratch

April 2026 – May 2026

- Developed a Multi-Layer Perceptron (MLP) with Backpropagation and a Single-Layer Perceptron using pure Python and NumPy, completely bypassing external ML frameworks like TensorFlow or PyTorch.
- Achieved 100% accuracy in character recognition on a custom dataset and logic gate classification, effectively demonstrating the XOR limitation of single-layer perceptrons.
- Built an interactive GUI using CustomTkinter for real-time model training and testing, and visualized decision boundaries using Matplotlib.

Experience

Agito Software & Consulting

Istanbul, Türkiye

Intern

July 2025 – September 2025

- Optimized SQL queries and designed relational database schemas, significantly contributing to data integrity and internal system performance improvements.

Zurich Insurance

Istanbul, Türkiye

End User Support Specialist

September 2024 – October 2024

- Delivered direct technical support and resolved complex troubleshooting requests, ensuring the seamless operation of end-user systems across the company.

Zurich Insurance

Istanbul, Türkiye

Intern

July 2024 – September 2024

- Diagnosed and resolved software and hardware issues, providing hands-on assistance to improve the overall employee user experience within the organization.

LC Waikiki Retail Services Inc.

Istanbul, Türkiye

Sales Assistant

December 2022 – May 2024

- Facilitated customer shopping experiences and maintained strict store organization, contributing to a smooth and efficient retail environment.

Skills & Interests

Technical: Python, Java, SQL, Docker, FastAPI, MLflow, Scikit-learn, flwr (Federated Learning), SQL Server, Git

Language: Turkish (Native), English (Fluent)

Interests: Artificial Intelligence, Machine Learning, Embedded Systems, AI Trend Analysis, Creative Writing